

PREMEDICATION IN ANAESTHESIA

Definition & goals

- **Premedication** = drugs given before anaesthesia to optimize comfort, reduce complications, improve outcomes.
- **Goals:** anxiolysis/sedation, haemodynamic stability, **reduce gastric volume/acidity**, decrease anaesthetic needs, facilitate induction + analgesia, **prevent PONV**.

Drug classes

Benzodiazepines, opioids, anticholinergics, antiemetics, H2 blockers/PPIs/antacids, alpha-2 agonists/beta-blockers, NSAIDs.

Anxiety management

- High incidence; anxiety → ↓satisfaction + ↑PONV + ↑acute pain. **Non-pharmacological:** rapport, explanation, empathy, reassurance, relaxation.

Benzodiazepines

- **Indications:** anxiolysis, sedation, **amnesia**, anticonvulsant, muscle relaxation (dose-dependent).
- **Mechanism:** enhance **GABA at GABA-A receptors** → ↑Cl⁻ influx → hyperpolarization.
- **Dosing:** lower in elderly, adjust for weight/liver/renal.
- **Midazolam:** 1–2mg IV / 7.5–15mg PO; onset 3–5min IV; short-acting (high lipid solubility); **very strong amnesia**.
- **Diazepam** (5–10mg, 2–4h, moderate amnesia), **Lorazepam** (1–2mg, long 6–8h, strong amnesia).
- **Side effects:** **respiratory depression (esp. with opioids)**, hypotension (↓SVR), paradoxical reactions, prolonged sedation, amnesia (problematic for instructions).
- **CI:** severe respiratory insufficiency, allergy. Caution: elderly, hepatic/renal, substance abuse. **Pregnancy: avoid**.
- **Paediatric: midazolam = gold standard** (PO 0.25–0.5mg/kg, intranasal 0.2–0.3mg/kg; multiple routes, rapid onset).

Analgesia (multimodal)

- Pre-emptive → prevent central sensitization, ↓opioid use, ↓postop pain.
- **Classes:** opioids, NSAIDs, paracetamol, gabapentinoids, alpha-2 agonists, NMDA antagonists, steroids, local anaesthetics.
- **Opioids:** fentanyl 1–2 mcg/kg IV (30min pre); morphine 0.1–0.2 mg/kg (severe, longer). **NSAIDs** (ketorolac, ibuprofen, diclofenac) — mild-moderate.
- **Paracetamol** 1g IV/PO. **Gabapentin** 300–600mg / **pregabalin** 75–150mg (block Ca channels → ↓neuronal excitability).
- **Ketamine** (NMDA) 0.1–0.5 mg/kg; **clonidine/dexmedetomidine** (alpha-2) = analgesia + sedation.

Other agents

- **Beta-blockers:** continue perioperatively if already on.
- **Anticholinergics** (atropine, glycopyrrolate, scopolamine): ↓secretions, prevent bradycardia.

- **Antiemetics (PONV): 5-HT₃ antagonists = gold standard** (ondansetron 4–8mg, granisetron 1mg); dopamine antagonist (**metoclopramide** — ↑gastric motility, ↓aspiration); steroids (dexamethasone).
- **Acid suppression:** H₂ blockers/PPIs (ranitidine, omeprazole) ↓acid + volume; **antacid (sodium citrate)** ↑pH but NOT volume (GERD, trauma, obstetrics).
- **Local anaesthetics:** block Na⁺ channels. Lidocaine 1–2% (blunts propofol pain), bupivacaine 0.25–0.5% (4–8h). **Paediatric: EMLA cream 30–60 min before cannulation.**

High-yield one-liners

- **Goals:** anxiolysis + ↓gastric volume/acid + ↓anaesthetic needs + prevent PONV.
- **Benzodiazepines = GABA-A; midazolam = strong amnesia + paediatric gold standard; SE = respiratory depression with opioids.**
- **5-HT₃ (ondansetron) = gold standard antiemetic; metoclopramide ↓aspiration risk.**
- **Sodium citrate (antacid) ↑pH not volume — obstetrics/trauma; EMLA for paediatric cannulation.**

PREMEDICATION IN ANAESTHESIA (Revision Table)

Bold = high-yield. Dr Gona Ahmed.

GOALS & DRUG CLASSES	
Topic	Key points
Goals	Anxiolysis/sedation, haemodynamic stability, ↓gastric volume/acidity, ↓anaesthetic needs, analgesia, prevent PONV
Classes	Benzodiazepines, opioids, anticholinergics, antiemetics, H2/PPI/antacid, alpha-2/beta-blocker, NSAIDs
BENZODIAZEPINES	
Aspect	Key points
Mechanism	Enhance GABA at GABA-A → ↑Cl ⁻ → hyperpolarization (sedation, amnesia, anxiolysis)
Midazolam	1–2mg IV; rapid, short; very strong amnesia; paediatric gold standard
Diazepam/Lorazepam	Diazepam (moderate amnesia 2–4h); lorazepam (strong, 6–8h)
Side effects	Respiratory depression (with opioids) , hypotension, paradoxical reaction; avoid in pregnancy
ANALGESIA & OTHER	
Class	Key points
Opioids	Fentanyl 1–2mcg/kg; morphine 0.1–0.2mg/kg (severe)
Non-opioid	Paracetamol 1g; gabapentinoids (Ca channel block) ; NSAIDs; ketamine (NMDA); clonidine/dexmedetomidine (alpha-2)
Anticholinergics	Atropine/glycopyrrolate/scopolamine — ↓secretions, prevent bradycardia
Antiemetics	5-HT3 (ondansetron) = gold standard ; metoclopramide (↓aspiration); dexamethasone
Acid suppression	H2/PPI (↓acid + volume); sodium citrate antacid (↑pH not volume) — obstetrics/trauma
Local anaesthetics	Na ⁺ block; lidocaine (blunt propofol pain), bupivacaine; EMLA for paediatric cannulation